

Family Name :

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Group :

Exercise 1 In this exercise, we want to study the regularity of the function defined by the following integral

$$F(x) = \int_1^{+\infty} \frac{\cos^2(xt) + \sin^2(t)}{t^2 \sqrt{1+t^3}} dt, \quad x \in [0, +\infty[.$$

1. Show that F is continuous on $[0, +\infty[$.

F is defined by an improper integral. (0.5)

We have : $f : \Delta = [1, +\infty[\times [0, +\infty[\longrightarrow \mathbb{R}$
(0.5) $(t, x) \longmapsto f(t, x) = \frac{\cos^2(xt) + \sin^2(t)}{t^2 \sqrt{1+t^3}}$

F is continuous on $[0, +\infty[\iff \begin{cases} \bullet f \text{ is continuous on } \Delta \\ \bullet \int_1^{+\infty} f(t, x) dt \text{ satisfies the dominated convergence criterion on } [0, +\infty[. \end{cases}$

1. f is continuous on Δ since it is composition, sum ratio of continuous functions on Δ . (1pt)

2. We have :

$$\forall (t, x) \in \Delta : |f(t, x)| = \left| \frac{\cos^2(xt) + \sin^2(t)}{t^2 \sqrt{1+t^3}} \right|$$

$$\leq \frac{\cos^2(xt)}{t^2 \sqrt{1+t^3}} + \frac{|\sin^2(t)|}{t^2 \sqrt{1+t^3}}$$

$$\leq \frac{2}{t^2 \sqrt{1+t^3}} = \varphi(t) \quad (1pt)$$

$$\text{and } \int_1^{+\infty} \psi(t) dt = \int_1^{+\infty} \frac{2}{b^2 \sqrt{1+t^3}} dt \sim \int_1^{+\infty} \frac{2}{t^{3/2}} dt$$

which is convergent. So $\int_1^{+\infty} f(t, x) dx$ satisfies

The dominated convergence criterion

Therefore F is continuous on $[a, +\infty[$.

2. Show that F is of class C^1 on $[0, +\infty[$.

$$F \text{ is of class } C^1 \text{ on } [0, +\infty[\Leftrightarrow \begin{cases} \cdot \exists x_0 \in [0, +\infty[\text{ s.t. } \int_1^{+\infty} f(t, x_0) dt < \infty \\ \cdot f \text{ continuous on } \Delta \\ \cdot \frac{\partial f}{\partial x} \text{ exists and continuous on } \Delta \\ \cdot \int_1^{+\infty} \frac{\partial f}{\partial x}(t, x) dt \text{ satisfies the dominated convergence criterion.} \end{cases}$$

1. Since $\int_1^{+\infty} f(t, x) dt$ satisfies the dominated convergence criterion on $[0, +\infty[$ so it is convergent on $[0, +\infty[$. Then there exists $x_0 \in [0, +\infty[$ s.t. $\int_1^{+\infty} f(t, x_0) dt < \infty$. (1pt)

2. f is continuous on Δ . (1pt)

3. $\frac{\partial f}{\partial x}$ exists on Δ and we have:

$$\frac{\partial f}{\partial x}(t, x) = \frac{-2 \sin(xt) \cos(xt)}{t \sqrt{1+t^3}} \quad (1pt)$$

$\frac{\partial f}{\partial x}$ is a continuous function on Δ since it is composition, sum, ratio of continuous functions on Δ . (1pt)

$$4. \left| \frac{\partial f}{\partial x}(t, x) \right| = \left| \frac{-2 \sin(xt) \cos(xt)}{t \sqrt{1+t^3}} \right| = \frac{2 |\sin(xt)| |\cos(xt)|}{t \sqrt{1+t^3}}$$

$$\leq \frac{2}{t \sqrt{1+t^3}} \leq \frac{2}{t^4} = \psi(t) \quad (1pt)$$

and $\int_{-\infty}^{+\infty} \psi(t) dt \in \mathcal{C}^1$ (spt)

So $\int_{-\infty}^{+\infty} \frac{\partial F}{\partial x}(t, x) dt$ satisfies the dominated

convergence criterion on $[0, +\infty[$

Therefore F is of class $\mathcal{C}^1$ on $[0, +\infty[$.

Exercise 2 Part I. In this part we want to find the extrema of the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x^2 + y^2 - 2x - 4y + 5$$

1. Find the critical points of f .

f is of class C^2 so $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exists and we have:

$$\nabla f(x, y) = \begin{pmatrix} 2x - 2 \\ 2y - 4 \end{pmatrix}$$

$$\nabla f(x, y) = 0_{\mathbb{R}^2} \Leftrightarrow \begin{cases} 2x - 2 = 0 \\ 2y - 4 = 0 \end{cases} \Rightarrow \begin{matrix} x = 1 \\ \text{and} \\ y = 2 \end{matrix}$$

So we have one critical point $A = (1, 2)$.

2. Find the nature of the critical points.

$$H_f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \forall (x, y) \in \mathbb{R}^2, \text{ and in particular}$$

$$H_f(1, 2) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\text{we have } \Delta_1 = |2| = 2 > 0$$

$$\text{and } \Delta_2 = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4 > 0$$

Then the point $A = (1, 2)$ realize a local minimum of f .

Part II. In this part we want to determine the extrema of the problem

$$\begin{cases} f(x, y) = x^2 + y^2 - 2x - 4y + 5 \\ \text{under the constraint} \\ x + y = 3 \end{cases}$$

1. show that there is no doubtful point of f under the constraint.

We set $g(x, y) = x + y - 3$

$$\nabla g(x, y) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \neq 0_{\mathbb{R}^2} \quad \forall (x, y) \in \mathbb{R}^2$$

(1pt)

then There is no doubtful points

2. Introduce the Lagrangian function L and find its critical points

$$L(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

(1pt)

then

$$L(x, y, \lambda) = x^2 + y^2 - 2x - 4y + 5 + \lambda(x + y - 3)$$

$$\nabla L(x, y, \lambda) = 0_{\mathbb{R}^3} \Leftrightarrow \begin{cases} 2x - 2 + \lambda = 0 & (1) \\ 2y - 4 + \lambda = 0 & (2) \\ x + y - 3 = 0 & (3) \end{cases}$$

(1pt)

from (1) and (2) we get

$$\lambda = 2 - 2x = 4 - 2y$$

$$\Rightarrow 1 - x = 2 - y$$

$$\Rightarrow y = x + 1 \quad (4)$$

from (3) after replacing (4) we get

$$x + x + 1 = 3 \Rightarrow 2x = 2 \Rightarrow x = 1$$

$$\text{and } y = x + 1 \Rightarrow y = 2$$

(4pt)

Then the critical point of α is $B = (1, 2)$.

3. Find the nature of each critical point of The Lagrangian

$$\begin{cases} f(1+h, 2+k) - f(1, 2) = (1+h)^2 + (2+k)^2 - 2(1+h) \\ \quad - 4(2+k) + 5 = 0 \\ (1+h) + (1+k) = 3 \end{cases}$$

$$\begin{cases} f(1+h, 2+k) - f(1, 2) = -h^2 + k^2 \\ k = -h \end{cases}$$

(SPT)

$$\begin{cases} f(1+h, 2+k) - f(1, 2) = 2h^2 > 0 \\ k = -h \end{cases}$$

Then The point $B = (1, 2)$ is a local min
of f under The constraint g .